

EUGENE L. BUCHANAN

Quality, Test & Controls Engineer

Apple Valley, CA · +1 (909) 545 5384 · eugene@serviceofothers.org · github.com/eugeneb50



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Exowatt Talent Network
Exowatt, Inc.
Via: jobs.lever.co/exowatt

Re: Quality, Reliability, Controls & Firmware, and Supply Chain Program — Exowatt Talent Network

Dear Exowatt Talent Network team,

I'm applying through the Exowatt Talent Network. I am a quality and controls engineer with thirty years across test engineering, product quality ownership, and thermal-energy systems, and I am relocating to Austin, TX or Miami, FL and available immediately. The four roles I am targeting are Quality Engineer, Reliability Engineer, Lead Controls & Firmware Engineer, and Technical Program Manager — Supply Chain & Manufacturing.

On quality and test ownership. At Knowledgecity I built the test automation function from nothing — recruited, hired, and trained the engineers, then built an end-to-end regression suite with health dashboards and alerting across complex deployment scenarios, so regressions surfaced on a dashboard rather than in a customer ticket. I set the quality bar across integrations, front end, back end, mobile, information security, performance, load, API, database, usability, accessibility, and localization including right-to-left Arabic. At IBM I drove WHQL certification testing for high-availability cluster servers, exercising RAID configurations under sustained high-load webserver traffic while injecting simulated drive failures, and proved failover held with no data or service lost across controller, array, and host failure. I have also worked blue-team incident forensics, taking each failure to root cause rather than symptom.

On controls and hardware debugging. At RealNetworks I tested decode stream optimization across architectures including ARM, programming directly on Psion 9210 Communicators running EPOC and Pocket PC — flashing builds to the device, capturing crash and memory-allocation logs off-target, and reproducing playback stalls under low-bandwidth and low-memory conditions. That isolated media pipeline defects that simulator-based testing had been passing, because the failure existed only on the device. More recently I delivered a SCADA and instrumentation upgrade for tank and pump sensors and relays, adding remote telemetry and automated control to district facilities, and sold and developed solar-fired VFD air-conditioning pumping projects at 30HP and above through concept, approval, and commissioning.

On solar design and permitting. I have worked in solar and renewable energy since 2004, and I produce permit-ready packages in CAD — plot plans, single-line electrical diagrams, equipment sizing, string impedance calculations, and NEC 690 compliance and safety calculations, plus rack load and footing calculations for site-specific structures. I coordinate licensed-engineer review and stamping, then run the package through city plan review to sign-off and permit to operate, and I now build those packages on AI-aided drafting software. I also design and fabricate custom mechanical parts where a catalog item does not fit. Enphase now ships a do-it-yourself permit package generator, which tells you where the industry thinks this is going. I build the same class of artifact by hand, and I can tell you exactly where a generator stops: custom mechanical parts that are not in a catalog, and sites where the impedance and footing calculations have to be right the first time. That work is what turns a solar thermal product into a permitted, interconnected, financeable site, and a modular unit system scaled to remote and international sites needs that pipeline to be cheap and fast or the deployment economics do not close. I have sold and integrated more than \$32 million in solar, wind, and geothermal air-conditioning installations while overseeing teams and vendors across multi-technology portfolios, developed solar-fired VFD air-conditioning pumping projects at 30HP and above through concept, approval, and commissioning, and built and ran a solar marketing company as CTO across sales, procurement, permitting, and finance. The

research I am proudest of is process stabilization, hot gas flow control, and anti-fouling for Fischer-Tropsch syngas — dirty, corrosive, high-temperature gas is a fouling problem, and it is the same fouling problem your heat exchangers live with.

On program and cost control. As president of a California water district I cut operating expenditures 53% on a \$16,027,777 annual budget while explicitly protecting capital projects, rebuilt district operations, delivered a SCADA and instrumentation upgrade for tank and pump sensors and relays with remote telemetry, and governed under the Brown Act through a term that included active litigation.

How I map to each role:

Quality Engineer	Built a test automation function from scratch. Owned acceptance criteria and quality metrics across integrations, front end, back end, mobile, security, and performance. Root-cause investigation and blue-team forensics. As a public officer, owned inspections, internal controls, and a documented public record.	Supplier quality — FAI, PPAP, SCAR, first-article inspection — and ISO 9001 QMS operation.
Reliability Engineer	Failure-injection test design under sustained load at IBM: RAID configurations with simulated drive failure, proving no data or service loss across controller, array, and host failover. Long practice taking field failures to root cause and feeding them back into design.	Accelerated life testing programs and Weibull life data analysis.
Lead Controls & Firmware Engineer	C and C++ on ARM: programmed directly on Psion 9210 devices, flashing builds and capturing off-target crash and memory logs. SCADA and instrumentation delivery for sensors, relays, and remote telemetry. VFD and motor control at 30HP+.	Current firmware practice — the device-level work is 1999–2001, and the controls work is supervisory rather than hands-on.
Technical Program Manager — Supply Chain	More than \$32M in solar, wind, and geothermal installations sold and integrated with vendor and contractor management. CAD permit packages — plot plans, single-line electrical, string impedance and NEC 690 safety calcs, rack load and footing calcs, engineer stamping, city plan review, permit to operate — plus custom mechanical part design. Delivered a \$16.0M operating program with a 53% reduction.	NPI and pilot-build management for manufactured hardware rather than site-built projects.

Column three is what I would want to prove in the first ninety days, not a claim of qualification.

Three of these four roles ask for a Bachelor's degree in engineering. I hold an Associate of Science in Mechanical Engineering from Victor Valley College, completed 1996, and what followed is thirty years of engineering work. I would rather raise that directly than have a resume screen decide it silently — if the degree requirement is firm, the roles where it is not the primary filter are the ones I can prove today.

I build quality and test programs, and I have spent twenty years turning heat and light into delivered hardware. I would welcome a technical conversation about the P3 system and where a test and controls organization would earn its keep in it.

Best regards,

Eugene L. Buchanan

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